~~Linear~~ Solving $(AX=0)$: pivot variables, special solutions

$$A=\begin{bmatrix}1&2&2&2\\2&4&6&8\\3&6&8&10\end{bmatrix}$$

$$\to\begin{bmatrix}1&2&2&2\\0&0&2&4\\0&0&2&4\end{bmatrix}$$

$$\to\begin{bmatrix}1&2&2&2\\0&0&2&4\\0&0&0&0\end{bmatrix}$$ = U

echelon

two pivots col 1, 2 free cols

rank = number of pivots = 2

$$\begin{bmatrix}0\\0\\0\end{bmatrix}x_1+2x_2+2x_3+2x_4=0$$

$$2x_3+4x_4=0$$

$$x=\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}\quad\begin{bmatrix}2\\0\\-2\\1\end{bmatrix}$$

$Ax=0,\ Ux=0$

pivots $=n-r$

# R = Reduced row echelon form

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

zeros above & below pivots

$$\begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \boxed{1} & 2 & 0 & -2 \\ 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\hookrightarrow = R = rref(A)$

## rref form:

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$I$ → pivot rows

$F$ ← $n-r$ free cols

$\begin{bmatrix} \text{pivot} \\ \text{cols} \end{bmatrix}$

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$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \\ 0 & 4 & 2 \\ 0 & 6 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x = c \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$